

Clarifying the “Boundary” from the Maximum Gap Method

You have a “Maximum Gap Method” that finds a *threshold* in your conditional probabilities

$$P(x + 1 \mid x)$$

by looking for the largest “drop” (negative jump) among consecutive probabilities. In your example, the largest down-step happened from

$$P(5 \mid 4) = 0.7897 \quad \text{to} \quad P(6 \mid 5) = 0.5502,$$

a drop of approximately -0.2394 . From that drop, your code computes a threshold,

$$\theta = 0.7897 + (-0.2394) \approx 0.5502.$$

That number, 0.5502, is what the code calls the “Maximum Gap Method” threshold.

1. What Exactly is the “Boundary”?

- **In Probability Terms:** The threshold $\theta = 0.5502$ is a dividing line on the probability scale. If $P(x + 1 \mid x)$ is *below* 0.5502, you might label it as “less likely to continue.” If $P(x + 1 \mid x)$ is *above* 0.5502, you might call it “more likely to continue.”
- **In Cluster-Size Terms:** Since the largest drop occurs between $x = 4$ and $x = 5$ (i.e. from $P(5 \mid 4) = 0.7897$ down to $P(6 \mid 5) = 0.5502$), the data suggests there’s a relatively sharp decrease in the probability of continuing the cluster after it reaches 5 pellets. Put another way:

$$P(5 \mid 4) \approx 0.79 \quad (\text{quite high}), \quad P(6 \mid 5) \approx 0.55 \quad (\text{lower but still above } 0.5).$$

So you can interpret the *boundary* as being around **cluster size 5**, because the probability of going from 5 to 6 happens to be your threshold of 0.5502. It is neither extremely high nor extremely low—it is simply the point where the maximum *drop* was detected.

2. Is It “Unlikely” or “Likely” After the 5th Pellet?

Both interpretations can arise, depending on how strictly you treat the threshold:

1. If your rule is “below $\theta = 0.5502$ means *stop the cluster*,” then 0.5502 is *exactly* on that boundary. A tiny change in decimal places (e.g. 0.55015 vs. 0.55025) could push it slightly over or under. In practice, you might label 0.55 as “borderline” or “marginally likely.”
2. If your rule is that you need $P(x+1 \mid x) > \theta$ by some margin (say strictly greater), then 0.5502 might *just barely* be included in the “continue” zone or *just barely* be excluded. This is a subtlety of any threshold-based classification.

Hence, saying “*it is unlikely to get more pellets after the 5th*” is *not* quite correct in absolute terms, because 0.55 is only slightly over one-half. It is better described as “the data show a moderate chance (roughly 55%) of continuing from 5 to 6, which is lower than going from 4 to 5 (about 79%), and that drop is the largest single jump in the curve, so the algorithm chose $\theta = 0.55$ as a boundary.”

3. Formula: $\theta = \max(P(x + 1 | x) - P(x | x - 1))$ and $\Delta P(x)$

Often, the code you are using for the “Maximum Gap” or “Maximum Drop” method does this:

1. **Compute $\Delta P(x)$ for each consecutive cluster-size x .** In your code, $\Delta P(x)$ is something like:

$$\Delta P(x) = P(x + 1 | x) - P(x | x - 1).$$

That is, it is the difference between consecutive conditional probabilities in the ordered list.

2. **Identify the index x for which the drop (the negative $\Delta P(x)$) is largest in magnitude.** Some implementations look for the *most negative* $\Delta P(x)$ and call it the “max gap.”

$$\text{“Max gap”} = \min_x(\Delta P(x)), \text{ because we want the largest negative jump.}$$

Then the code sets

$$\theta = [P(x | x - 1)] + [\Delta P(x)].$$

Or sometimes, the method is a bit different (they might look for the largest ratio, or they might do an alternative offset). Implementation details vary.

3. **Interpret θ as a cutoff** to classify probabilities:

$$P(x + 1 | x) \begin{cases} > \theta & \text{“more likely to continue”} \\ \leq \theta & \text{“less likely to continue”} \end{cases}$$

So the expression “ $\theta = \max(P(x + 1 | x) - P(x | x - 1))$ ” is a short way of saying: *we search for the maximum difference in consecutive probabilities (i.e., the biggest jump), and we use that to define or compute θ .*

Note that in practice, many labs or researchers will *manually* review these probabilities to see if the largest jump is truly a good dividing line or if some other threshold (like 0.70) makes more sense for the particular behavioral or biological context.

4. Bottom Line

- **The boundary is around cluster size 5** because the largest downward drop is between $P(5 | 4) = 0.7897$ and $P(6 | 5) = 0.5502$.
- **That gives a threshold of about 0.55.**
- *Does it mean “unlikely” to continue?* It depends on your classification rule. 0.55 is only slightly above 0.50, so it’s *moderately* likely in everyday terms, but it’s still a sizable drop from 0.79.

- **The formula with ΔP** picks out where consecutive probabilities differ the most. The “max” is the single biggest negative gap, and ΔP is simply the difference $P(x + 1 \mid x) - P(x \mid x - 1)$ at each step.

In other words, once the cluster reaches 5 pellets, the probability of adding a 6th pellet is 0.55, which is noticeably lower than the 0.79 chance of going from 4 to 5. That *drop* is the largest among consecutive sizes, so the maximum gap logic designates ≈ 0.55 as the threshold (your “boundary”). Whether you label 0.55 as “likely” or “not likely” depends on your own behavioral criteria, but mathematically, the method is simply capturing that big downward jump.